

PUGET SOUND RESTORATION FUND · UNIVERSITY OF WASHINGTON · NOAA
NWFSC

Conservation aquaculture maintains genetic composition in hatchery- produced Olympia oysters

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BACKGROUND

A native oyster reduced to remnants

~4%

of historic dense core Olympia oyster populations
populations remained in Puget Sound by the early
the early 2000s

Ostrea lurida is the only oyster native to the US west coast, once forming
beds from Southeast Alaska to Baja California.

Overharvest, sedimentation and pollution drove a range-wide collapse by
collapse by the early 1900s.

Since 2014, PSRF, NOAA and WDFW have produced restoration seed at
the Kenneth K. Chew Center.

WHY GENETICS MATTERS HERE

Puget Sound oysters are not one population

Microsatellites (Stick 2012)

75% of genetic variation partitioned among four sub-basins: North, Central, South Sound and Hood Canal

Genome-wide SNPs (Silliman 2019)

Strong neutral structure across the range, with putatively adaptive genes for development, metabolism and immunity

Ocean acidification (Spencer et al. 2020, 2023)

Populations differ in transcriptional and phenotypic responses, with evidence of transgenerational plasticity

Transplants and expression (Heare et al. 2017, 2018)

Basin-level differences in growth, survival, reproduction and stress response

Risk: hatchery seed could erode diversity or blur these distinct gene pools.

Seven practices designed to protect wild diversity

1

Large broodstock

Minimum of 1,200 adults

2

Local source

Matched to the restoration site's genetic stock

3

Wild origin

No hatchery-born parents

4

Annual refresh

New broodstock every year

5

Mass spawning

Volitional, no controlled crosses
crosses

6

Long collection

Larvae collected for at least 4
weeks

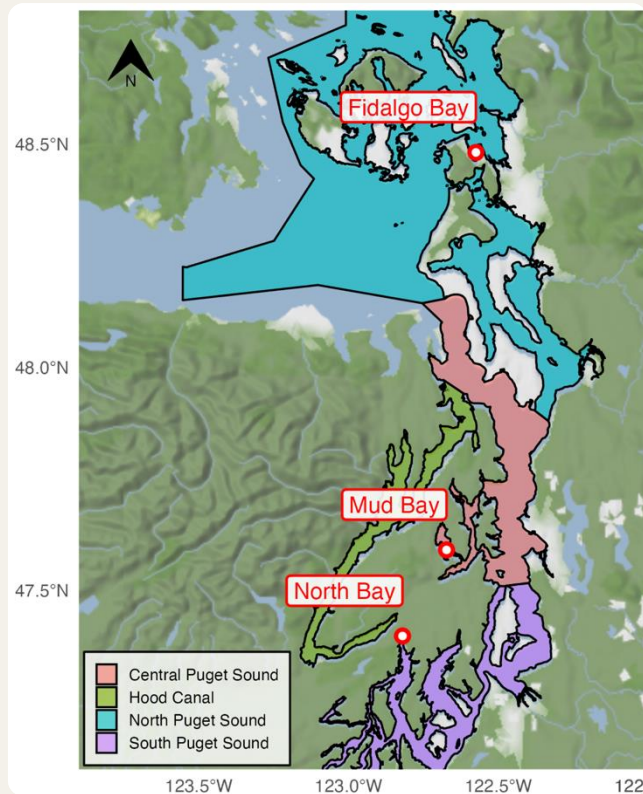
7

Even culling

Spread across the whole spawning
spawning period

STUDY DESIGN

Three source populations, hatchery vs wild



Source	Wild adults	Hatchery seed
North · Fidalgo Bay	2018	2017
Central · Mud Bay	2017	2018
South · North Bay	2018	2018

96 oysters per group (NW, NH, CW, CH, SW, SH); seed sampled at 5 to 10 mm before outplant. 23 batch-linked loci removed (NH library prepared separately during COVID-19).

561

oysters retained of 576

2,342

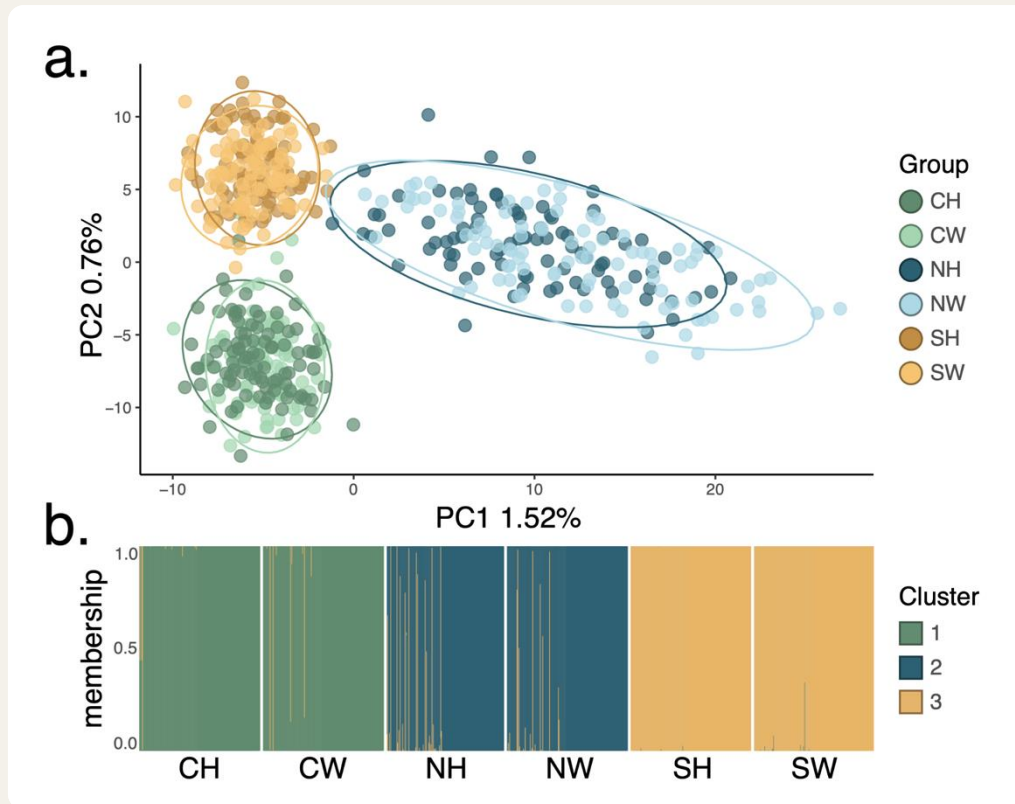
filtered SNPs

RAD

BestRAD, Sbfl, NovaSeq 150 PE

RESULT 1 · POPULATION STRUCTURE

Oysters cluster by source, not by rearing



Cluster 1: 185 of 190 Central

93 hatchery · 92 wild

Cluster 2: 168 of 184 North

79 hatchery · 89 wild

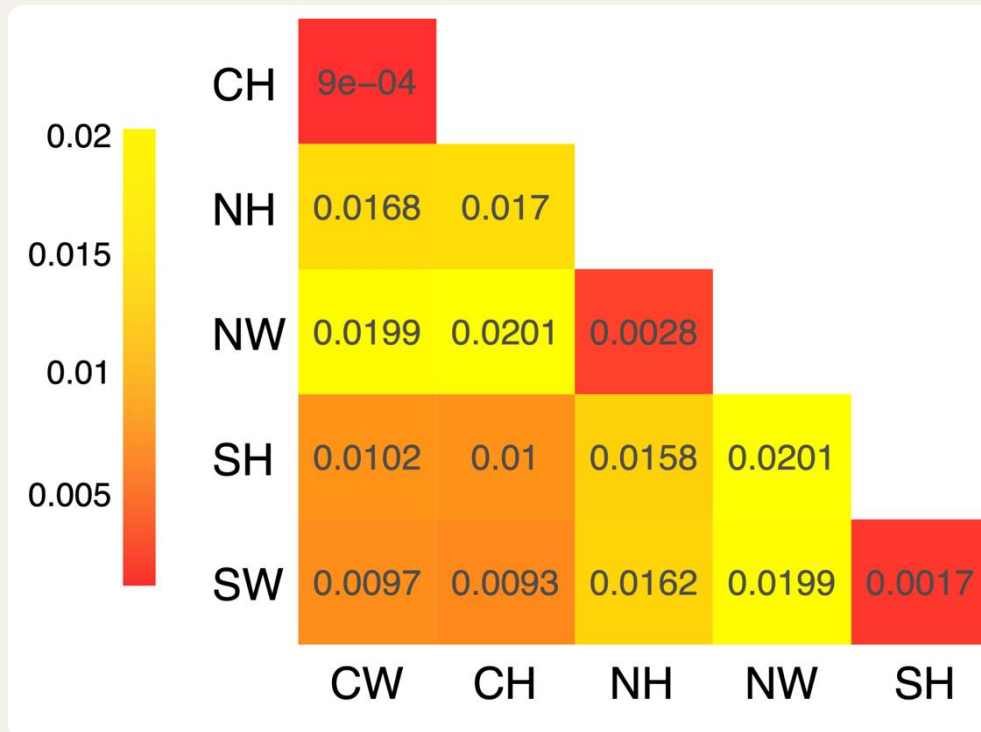
Cluster 3: all 187 South

Plus 21 North and Central individuals

No hatchery cluster: hatchery and wild intermix within every source.

RESULT 2 · DIFFERENTIATION

Each hatchery group is closest to its own wild source



0.0009 to 0.0028

Each hatchery group vs its own wild source: the three lowest values

~0.01

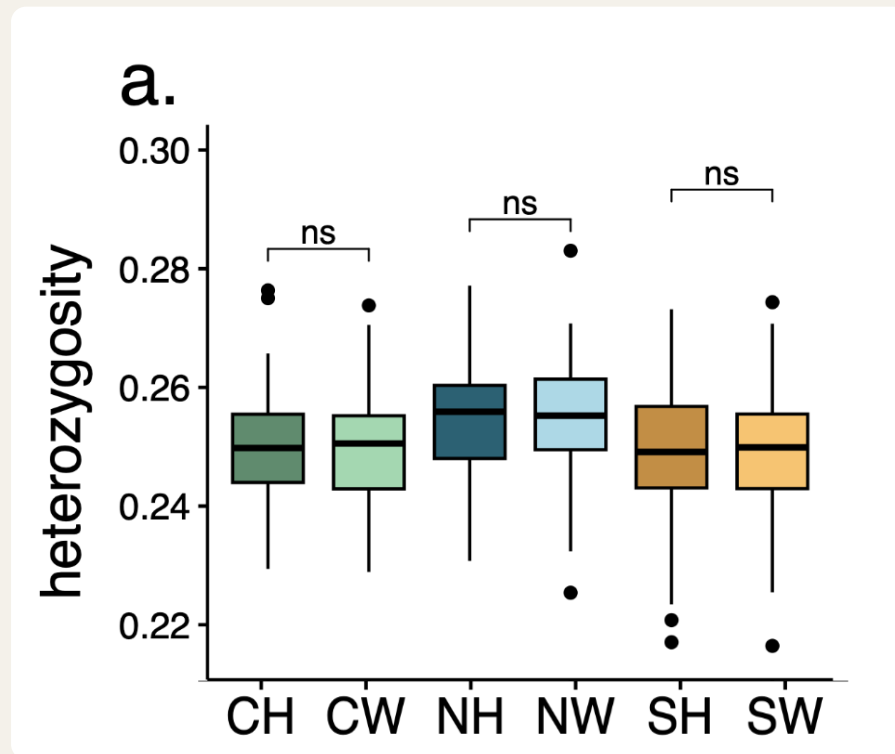
Central vs South, regardless of rearing

0.0158 to 0.0201

North vs the other sources

RESULT 3 · GENETIC DIVERSITY

No loss of heterozygosity, no rise in inbreeding



Heterozygosity (H_o)

Wild

0.2514

Hatchery

0.2509

Inbreeding (F_{is})

Wild

0.2361

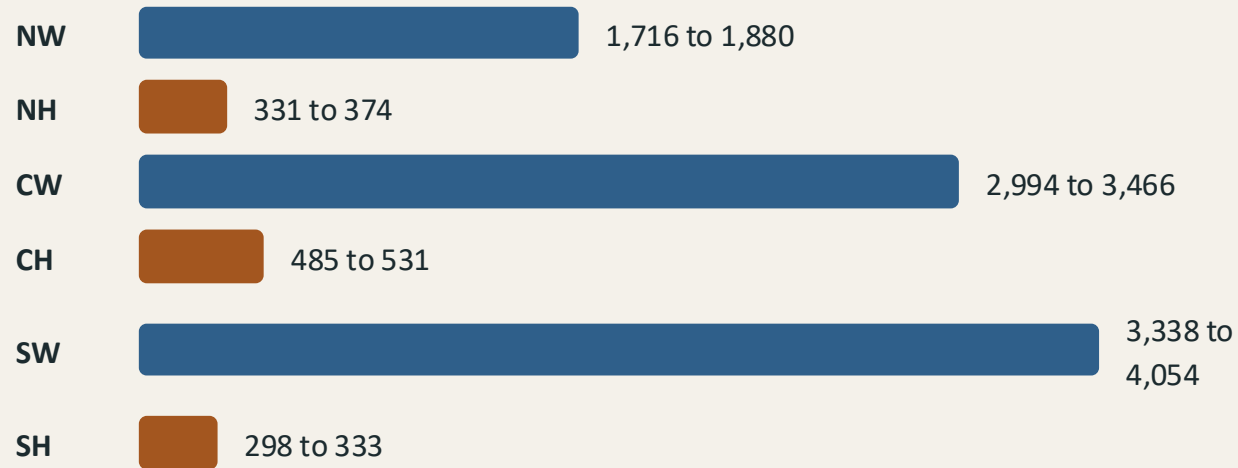
Hatchery

0.2365

Neither difference significant (panels a, d). North is highest in H_o and lowest in F_{is} in lowest in F_{is} in both wild and hatchery groups (b, c, e, f).

RESULT 4 · EFFECTIVE POPULATION SIZE

Hatchery Ne is lower, but well above a family sweep



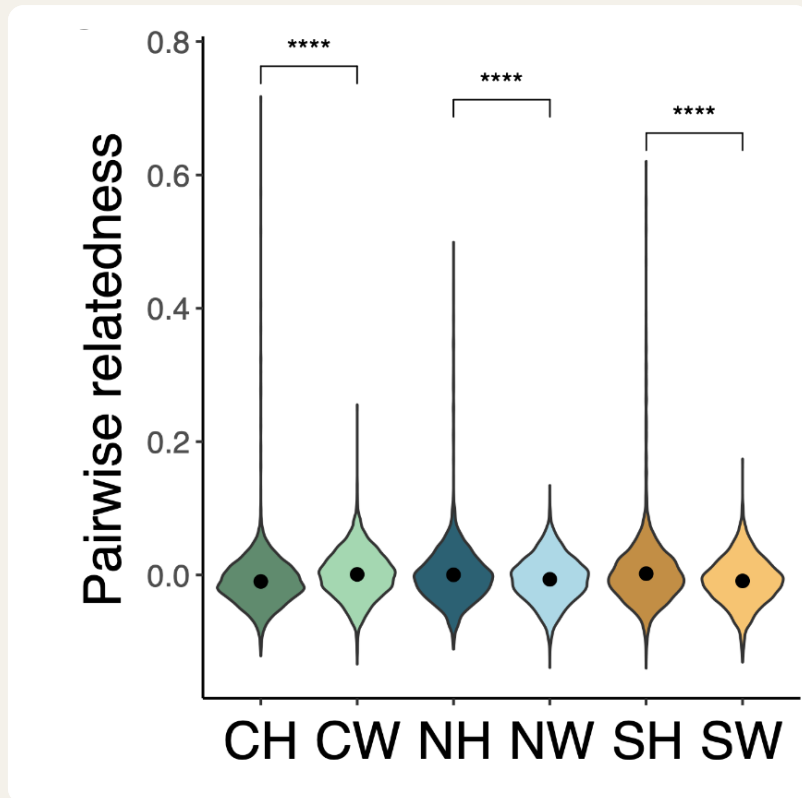
LD Ne, NeEstimator v2.1 (Pcrit 0.05 to 0.01). Blue = wild, orange = hatchery.

Wild Ne in the low to mid thousands resembles less impacted eastern oyster populations.

Hatchery Ne of roughly 300 to 530 is comparable to natural and restored Chesapeake Bay reefs (~70 to 500).

RESULT 5 • RELATEDNESS

Siblings appear in hatchery seed, but rarely



0.03%

of wild pairs were siblings; no full siblings

1.07%

of hatchery pairs were siblings

Hatchery group	Full sibs	Half sibs
SH	0.16%	1.42%
NH	0.11%	0.89%
CH	0.11%	0.51%

Far from sweepstakes reproduction.

LIMITATIONS AND NEXT STEPS

What this study cannot tell us yet

Adaptive variation

2,342 neutral SNPs resolve structure and kinship but not loci under selection. Whole-genome sequencing could test whether adaptive alleles are retained.

Field performance

Genetic similarity is not fitness. In situ comparisons of hatchery and wild survival, growth and reproduction are the real test.

Scalability

1,200+ broodstock is hard for most programs, and N_e tracks broodstock number in oysters (Hornick and Plough 2022).

TAKEAWAYS

A protocol that keeps wild genetics intact

- 1 Hatchery seed kept its source population's identity, and wild structure was preserved.
- 2 Diversity held steady; only N_e and sibling frequency shifted, and modestly.
- 3 Large wild broodstock, local sourcing and long larval collection are the levers.

Thank you

ACKNOWLEDGMENTS

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CODE AND DATA

github.com/evelynabbott/olympia_oyster_RAD

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